

Wireless Power Consortium

Aimed at establishing common standards and interfaces for wireless power, it includes Nokia, Philips, RIM, and strangely enough, battery manufacturer Energizer

magnetic energy interferes with some of the communication features in the device, such as GSM or radio.

This kind of resonant coupling, is a great compromise between efficiency and mobility. Other kinds of wireless electricity technologies either need the devices to be too close to the source, or cannot charge up any device in a reasonable amount of time. The inductive coupling approach also is the lowest cost solution, and can be cheaply deployed in a number of ways. The coil to charge a device can be either embedded within the device itself, replacing the battery unit, or added to the back of a device as an additional sleeve, which is connected to the power unit of the device. The sleeve approach is great for transition between a wired domestic, office and public environments, to a wireless one. The device can be charged using traditional methods where wireless chargers are not available. The charging can be done in the form of mats, that can charge multiple devices at once. These sources



Image credit: Powercast
A Powerharvester module, is tiny enough to be embedded almost anywhere

can be embedded, within say the tables of cafes or the walls of homes.

Since the technology has been around for a while, and is a low cost solution, why are we not seeing more wireless devices around? When asked this question, the WiPower team told us that "The major hurdle to mass adoption has been the difficulty and cost of integrating wireless power technology into consumer electronic products. Large OEMs are very interested in the technology, but have generally found it too complex and expensive to replicate in high volume manufacturing environments. " The initial cost of such devices



Uniden Power Pad Australia

would be high, and it is an overhead that consumers may not be willing to pay extra for. For example, if a manufacturer comes out with a camera that charges wirelessly, will the buyer pay extra for a mat that can charge the camera, and maybe other devices from manufacturers who follow the same standard? It is easier to market a wireless television, that does not need to be tethered to a power source.

There is currently a gap between wireless technologies, and its implementation, as manufacturers will have to rethink the way they make their devices. "WiPower has attempted to bridge this gap by providing companies with a wireless power Design Kit. The Design Kit allows engineers to explore WiPower's technology, and discover how simple and cost-effective it is to design into a variety of applications. With WiPower's wireless power system, consumers will be able to charge multiple devices in any position on a single charging pad. The technology is also easily integrated into furniture so that people will be able to enjoy clean, wireless spaces at home, the office, etc." The WiPower design specifications are available to manufacturers, and orders can be placed for the transmitter and receiver elements to integrate into any manufacturing process. The receiver units are designed in such a way, that they can be potentially retro-fitted into existing consumer electronic products. One of WiPower's greatest innovations is in the charging pads. The pads are

configured in a way that they can calculate the amount of charge necessary for the device, and relay only that much power to the receiver.

Long range wireless power

The technologies for long-range wireless electricity transmission, a possibly more exciting domain, include radio waves, microwaves, and even lasers. The technologies are not that difficult to decipher or understand. For example, to use lasers to provide wireless power, you need to simply point a sufficiently powerful laser at a solar panel. It is easy to envision a network of lasers instead of wires, but there is the question of efficiency. While in controlled spaces, the efficiency is pretty good, over a large grid or deployment on a large scale is not yet practical. However, a small amount of charge can be "cast" over a large area, powering up devices that are willing to accept that charge.

Wirelessly relaying small, but steady amounts of power have their own

unique set of advantages. Powercast is a company that specialises in something known as "trickle power", a model of wireless power transmission that allows power to be relayed over radio waves in small quantities. They make Powerharvester Receivers, that can "harvest" electricity

and power devices, and Powercaster Transmitters, that are sources of power for the receivers. The use of Powercast for consumer technologies is very limited. According to Powercast, "Our technology provides micro-power and will not be suitable for most consumer devices, and especially devices like mobile phones that require fast recharging. Wireless power for consumer devices will primarily be based on induction and will work for higher power, short range applications." Powercast has demoed a few trickle power consumer applications though, including LED Christmas lights and remote controls that charge up wirelessly.

